CS6600 Assignment 1: Cache and Memory Hierarchy Simulator

Shubham Jadhav (CS24S009)

1 Simulator and validation

The simulator (`cache_sim.cpp`) is a single generic `Cache` class that is instantiated as L1 and, optionally, L2. Each set is kept in MRU→LRU order. L1 owns an optional fully associative victim cache (VC). Writes are write-back/write-allocate. A miss on a full L1 set first searches the VC (a *swap request*), and a hit swaps the two blocks. On a VC miss the L1 victim enters the VC, a dirty block pushed out of the VC is written back, and the missing block is read from the next level. L2 is non-inclusive. CACTI supplies hit time, energy and area (a VC that CACTI cannot model uses $HT_{VC} = 0.2$ ns).

All 8 reference outputs match with `diff -iw`: raw statistics, the contents of L1, VC and L2 in MRU→LRU order, AAT, EDP and area, with EDP matching digit for digit. A further 20 outputs (L1 from 8 KB to 1 MB, associativity up to 512, block sizes up to 2 KB, VC with 4–14 blocks, L2 up to 8 MB) also match for every line they contain. All experiments below use the provided `gcc` trace and are produced by `run_experiments.py`.

CACTI limits. CACTI cannot model some very small or highly associative arrays (for example a 1 KB or 2 KB 4-way cache, a 2 KB 8-way cache or a 512 KB 2-way cache with 32 B blocks). For those configurations AAT, EDP and area are undefined, so they appear as gaps in the plots; miss rates are unaffected.

2 L1 size and associativity

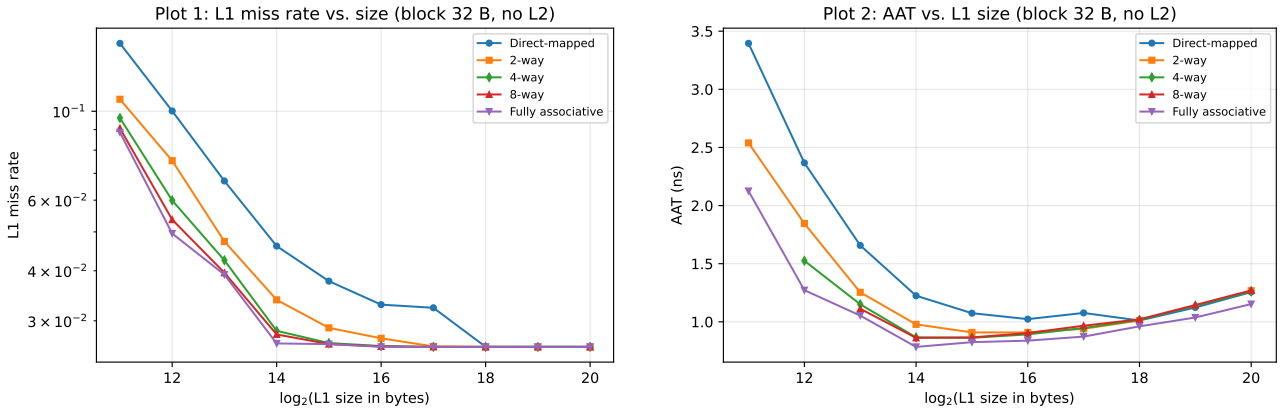


Figure 1: Plot 1 (L1 miss rate, log scale) and Plot 2 (AAT) versus L1 size for block size 32 B and no L2.

Plot 1: trends. Miss rate falls steadily with size and with associativity and then flattens. A direct-mapped cache falls from 14.8% at 2 KB to 3.2% at 128 KB. 4-way and 8-way caches reach the floor near 32 KB (2.64% and 2.62%), and every curve reaches 2.58% by 256 KB. Associativity helps most

for small caches: at 2 KB, going from direct-mapped to 8-way cuts the miss rate by about 40% (14.8% to 9.1%), whereas at 128 KB the 2-way, 4-way and 8-way caches are indistinguishable.

Compulsory miss rate. The floor is the miss rate of a cache large enough to hold the whole footprint: 2.58% (identical at 256 KB and 1 MB, fully associative).

Conflict miss rate (miss rate of the A-way cache minus that of the fully associative cache of the same size, in percentage points):

Size	2 KB	4 KB	8 KB	16 KB	32 KB	64 KB	128 KB
Direct-mapped	5.91	5.06	2.79	1.98	1.14	0.71	0.65
2-way	1.85	2.57	0.82	0.75	0.26	0.13	0.01
4-way	0.76	1.04	0.33	0.20	0.02	0.01	0.00
8-way	0.21	0.41	0.04	0.14	0.00	0.01	0.00

The remaining part of the fully associative miss rate above 2.58% is the capacity miss rate (6.3 points at 2 KB, 2.4 at 4 KB, 1.3 at 8 KB, nearly 0 from 16 KB). A direct-mapped cache still has 0.65 points of conflict misses at 128 KB, so conflicts outlive capacity misses.

Plot 2: best AAT without L2. Hit time rises with size and associativity, so AAT has an interior minimum: 4-way and 8-way curves bottom out at 0.86 ns (16–32 KB) and rise again to 1.27 ns at 1 MB, where the miss rate is already at its floor. The lowest AAT of all is the **16 KB fully associative** cache at 0.784 ns (the CACTI model of a small CAM is fast), with EDP 4.6×10^8 and area 0.063 mm^2. The best set-associative choices are 16 KB 8-way and 32 KB 4-way, both at 0.862 ns; the 16 KB 8-way point has EDP 1.46×10^8 and area 0.121 mm^2.

3 Adding an L2

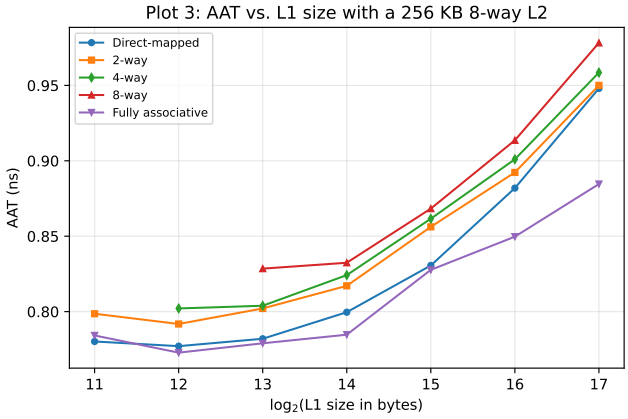


Figure 2: Plot 3: AAT versus L1 size with a 256 KB 8-way L2.

With the L2 the miss penalty seen by L1 collapses from about 22 ns to $HT_{L2} + MR_{L2} \times 22$ ns, so a cheap, fast L1 wins and the best L1 shrinks to 4 KB. The best AAT is **0.773 ns with a 4 KB fully associative L1** (0.777 ns for the 4 KB direct-mapped L1; 0.779 ns for 8 KB fully associative). Compared with the best L1-only AAT this is a 1.5% improvement over 0.784 ns (fully associative) and

a 9.9% improvement over 0.862 ns when only set-associative L1 caches are allowed (0.777 ns, 4 KB direct-mapped).

EDP and area. With the 4 KB direct-mapped L1 + L2 the EDP is 6.8×10^7 versus 1.46×10^8 for the best L1-only set-associative design (about 2.2× better), and it is about 7× better than the fully associative L1-only design (4.6×10^8). The price is area: 1.20 mm^2 versus 0.12 mm^2 (about 10×) because the 256 KB L2 dominates. The L2 buys energy by filtering DRAM traffic and costs silicon.

4 L1 size and block size

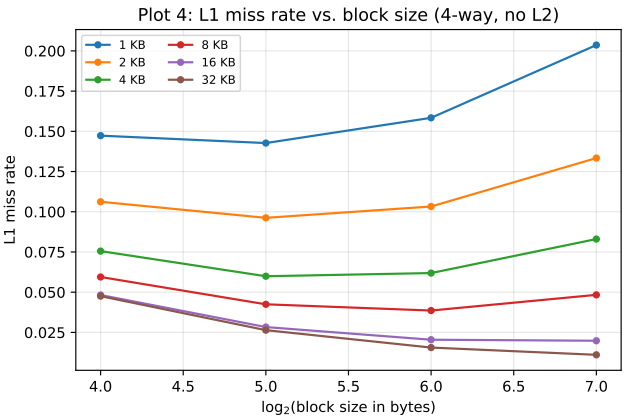


Figure 3: Plot 4: L1 miss rate versus block size for a 4-way L1, no L2.

The spatial-locality versus pollution trade-off is clear and depends on cache size. For 1–4 KB caches the miss rate is lowest at 32 B and rises sharply for 128 B (1 KB: 14.3% at 32 B, 20.4% at 128 B), because few large blocks mean few distinct blocks: pollution and conflicts dominate. For an 8 KB cache the minimum moves to 64 B (3.86%, versus 4.25% at 32 B and 4.83% at 128 B). For 16 KB and 32 KB caches larger blocks keep helping (32 KB: 4.75, 2.64, 1.56, 1.11% for 16, 32, 64, 128 B). The **optimal block size for an 8 KB 4-way L1 is 64 B** by miss rate; by AAT also 64 B (1.139 ns versus 1.151 ns at 32 B). A 16 B block is never best: its miss rate is highest for ≥8 KB caches.

5 L1 + L2 co-optimisation

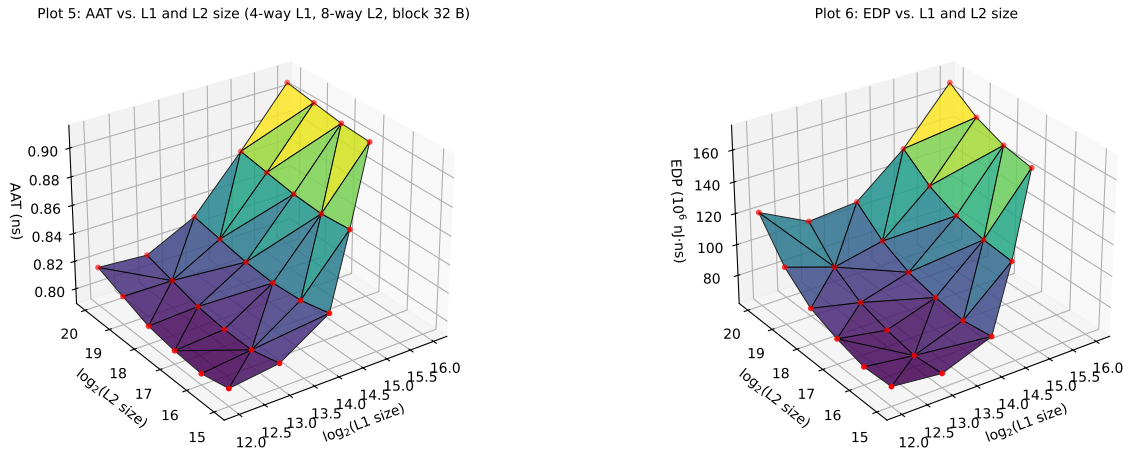


Figure 4: Plot 5 (AAT) and Plot 6 (EDP) for a 4-way L1 and 8-way L2, block 32 B, only points with L1 smaller than L2. CACTI cannot model a 1 KB or 2 KB 4-way L1 (and the 32 KB L1 with a 32 KB L2 is excluded), so those points are missing.

Best AAT. The best AAT is **0.797 ns with a 4 KB L1 and a 64 KB L2** (area 0.386 mm^2). The surface is almost flat in L2 size (0.797–0.82 ns for the 4 KB L1) and rises with L1 size, because the L1 hit time is paid on every access while a bigger L2 only helps the few L1 misses.

Smaller design within 5%. A 4 KB L1 with a **32 KB L2** has AAT 0.802 ns (0.6% above the best) with area 0.285 mm^2, 26% smaller; an 8 KB L1 with a 32 KB L2 (0.806 ns, +1.1%, 0.309 mm^2) also qualifies.

Best EDP (Plot 6). The best EDP is 6.86×10^7 **with an 8 KB L1 and a 64 KB L2** (AAT 0.801 ns, area 0.410 mm^2). EDP grows quickly with L2 size because the energy of every L2 access grows with its size, so energy favours small L2s even more than AAT does.

6 Victim cache

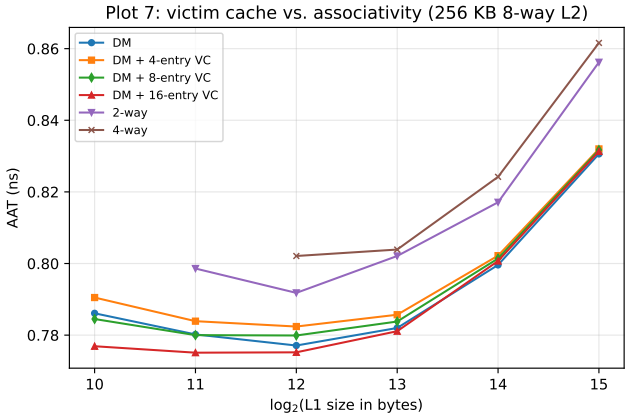


Figure 5: Plot 7: AAT versus L1 size with a 256 KB 8-way L2. Missing points are CACTI failures for the 1 KB and 2 KB 4-way cache and the 1 KB 2-way cache.

VC versus 2-way. For small direct-mapped caches a victim cache does better than doubling the associativity: at 2 KB a direct-mapped L1 with a 16-entry VC has AAT 0.775 ns against 0.780 ns for direct-mapped and 0.799 ns for 2-way; at 4 KB it is 0.775 versus 0.777 and 0.792. The 2-way cache removes the same conflicts but pays a longer hit time on every access, while the VC is probed only on a miss in a full set (swap request rate 0.15 at 2 KB, 0.10 at 4 KB and 0.03 at 32 KB). For 8 KB and larger the VC gains vanish (16 KB: 0.8007 with a 16-entry VC versus 0.7996 without; 32 KB: 0.8313 versus 0.8306 for direct-mapped), because the conflict misses it catches are few and the extra swap-request time and energy outweigh them.

Best configuration. Among the Plot 7 configurations the best AAT is **a direct-mapped L1 of 2 KB or 4 KB with a 16-entry VC, 0.775 ns** (0.7751 and 0.7752 ns). Smaller VCs help less (4 and 8 entries give 0.784 and 0.780 ns for the 2 KB cache).

Smaller area within 5%. Almost all of the area is the 256 KB L2 (1.2 mm^2). Replacing it by a 32 KB 8-way L2 gives a **4 KB direct-mapped L1 + 16-entry VC + 32 KB L2** with AAT 0.774 ns (slightly *better*) and area 0.285 mm^2 (76% smaller); with a 2 KB L1 the AAT is 0.770 ns at 0.264 mm^2.